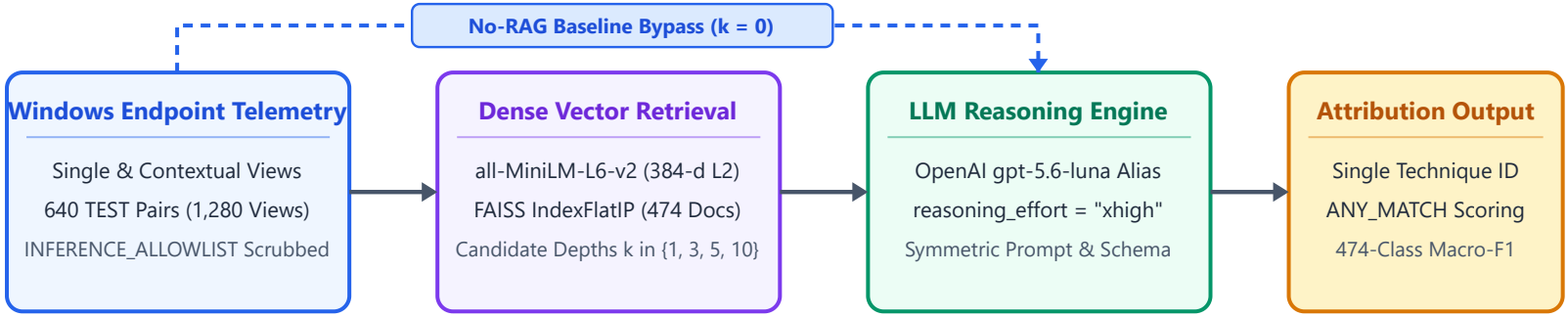
Figure 1: RAG2ATT&CK Dual-View Attribution Architecture

Controlled evaluation pipeline across 5 conditions (N = 1,280 TEST views; 718 scorable mapped-positive views)



Frozen Scientific Protocol v1.1 & Financial Ledger Governance (Decisions D1–D7)

- **D1: RECORD_ONLY Prompt & Output Policy**
  Raw model completions recorded verbatim with secret scrubbing
  Byte-identical prompt template across No-RAG and RAG
- **D2a–D2j: Scoring, Universe & NULL Semantics**
  ANY_MATCH accuracy on 718 mapped views (311 ambig / 251 unmapped excl.)
  474-class Windows universe (D2d); zero denominators export as null (D2j)
- **D3 & D4: Provenance & Sequential Execution**
  Sequential execution (concurrency = 1); fixed model alias & settings
  Timestamped response IDs retained (backend version/fingerprints null)
- **D5: Financial Guard & Worst-Case Reservation**
  USD 19.99 hard study ceiling; USD 0.53974560 pre-dispatch attempt hold
  Missing usage receipts retain full worst-case hold in settled ledger
- **D6 & D7: Synthetic Paired Benchmark Boundary**
  Task 2 forensic stop on Mendeley v3 (15,713 discrepancies); T15 deferred
  All claims strictly bounded to synthetic-paired-v1 (640 TEST pairs)
- **D2h & D2i: Retrieval Hit vs. Recall & Failure Axes**
  Hit@k ($Y_i \cap R_{i,k} \neq \emptyset$) distinguished from multi-label Macro Recall@k
  5 non-mutually-exclusive failure axes evaluated without forced causality